

Sometimes, code is just not worth the trouble of deduplicating. Compare the first code example in this section to the most recent one. Although the duplicate code is longer, it's simple and straightforward. The deduplicated example does the same thing but involves a loop, a new `timeOfDay` loop variable, and a new function with a parameter that is also named `timeOfDay`.

Duplicate code is a code smell because it makes your code harder to change consistently. If several duplicates are in your program, the solution is to place code inside a function or loop so it appears only once.

Magic Numbers

It's no surprise that programming involves numbers. But some of the numbers that appear in your source code can confuse other programmers (or you a couple weeks after writing them). For example, consider the number `604800` in the following line:

```
expiration = time.time() + 604800
```

The `time.time()` function returns an integer representing the current time. We can assume that the `expiration` variable will represent some point 604,800 seconds into the future. But `604800` is rather mysterious: what's the significance of this expiration date? A comment can help clarify:

```
expiration = time.time() + 604800 # Expire in one week.
```

This is a good solution, but an even better one is to replace these “magic” numbers with constants. *Constants* are variables whose names are written in uppercase letters to indicate that their values shouldn't change after their initial assignment. Usually, constants are defined as global variables at the top of the source code file:

```
# Set up constants for different time amounts:
SECONDS_PER_MINUTE = 60
SECONDS_PER_HOUR   = 60 * SECONDS_PER_MINUTE
SECONDS_PER_DAY     = 24 * SECONDS_PER_HOUR
SECONDS_PER_WEEK    = 7  * SECONDS_PER_DAY
```
